

```

Command Prompt - mysql -u x + v

mysql> create database dbms_lab;
Query OK, 1 row affected (0.05 sec)

mysql> use dbms_lab;
Database changed
mysql> create table roles(role_id int primary key,role_name varchar(50));
Query OK, 0 rows affected (0.10 sec)

mysql> create table users(user_id int primary key,name varchar(50),email varchar(50),role_id int);
Query OK, 0 rows affected (0.04 sec)

mysql> create table departments(dept_id int primary key, dept_name varchar(50));
Query OK, 0 rows affected (0.04 sec)

mysql> create table employees(emp_id int primary key,emp_name varchar(50),dept_id int,manager_id int,salary decimal(10,2));
Query OK, 0 rows affected (0.06 sec)

mysql> create table customers(customer_id int primary key,customer_name varchar(50),city varchar(50));
Query OK, 0 rows affected (0.03 sec)

mysql> create table orders(order_id int primary key,customer_id int,order_date date,total_amount decimal(10,2));
Query OK, 0 rows affected (0.04 sec)

mysql> create table accounts(
-> account_no int primary key,
-> holder_name varchar(50), balance decimal(10,2));
Query OK, 0 rows affected (0.03 sec)

```

```

Query OK, 0 rows affected (0.03 sec)

mysql> show tables;
+-----+
| Tables_in_dbms_lab |
+-----+
| accounts            |
| customers            |
| departments          |
| employees            |
| orders              |
| roles               |
| users               |
+-----+
7 rows in set (0.01 sec)

mysql> insert into users values(101,'Bhavya','bhavya@gmail.com',1),(102,'Priya','priya@gmail.com',2),
-> (103,'Arun','arun@gmail.com',3);
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO roles VALUES
-> (1,'Admin'),
-> (2,'Manager'),
-> (3,'Employee');
ERROR 2013 (HY000): Lost connection to MySQL server during query
No connection. Trying to reconnect...
Connection id: 17
Current database: dbms_lab

Query OK, 3 rows affected (0.09 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> insert into departments values (10,'CSE'), (20,'ECE'), (30,'EEE');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO employees VALUES
```

```
-> (1, 'Ramesh', 10, NULL, 80000),
```

```
-> (2, 'Suresh', 10, 1, 60000),
```

```
-> (3, 'Mahesh', 20, 1, 45000),
```

```
-> (4, 'Kiran', 30, 2, 55000);
```

```
Query OK, 4 rows affected (0.01 sec)
```

```
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO customers VALUES
```

```
-> (201, 'Rahul', 'Hyderabad'),
```

```
-> (202, 'Anita', 'Delhi'),
```

```
-> (203, 'Kiran', 'Mumbai');
```

```
Query OK, 3 rows affected (0.01 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO orders VALUES
```

```
-> (1001, 201, '2026-08-01', 2500.00),
```

```
-> (1002, 202, '2026-08-02', 3200.00),
```

```
-> (1003, 201, '2026-08-03', 1500.00),
```

```
-> (1004, 203, '2026-08-04', 5000.00);
```

```
Query OK, 4 rows affected (0.01 sec)
```

```
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO accounts VALUES
```

```
-> (101, 'Bhavya', 50000),
```

```
-> (102, 'Priya', 30000),
```

```
-> (103, 'Arun', 45000);
```

```
Query OK, 3 rows affected (0.01 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from roles;

role_id	role_name
1	Admin
2	Manager
3	Employee

3 rows in set (0.00 sec)

mysql> select * from users;

user_id	name	email	role_id
101	Bhavya	bhavya@gmail.com	1
102	Priya	priya@gmail.com	2
103	Arun	arun@gmail.com	3

3 rows in set (0.00 sec)

mysql> select * from departments;

dept_id	dept_name
10	CSE
20	ECE
30	EEE

3 rows in set (0.00 sec)

mysql> select * from employess;

ERROR 1146 (42S02): Table 'dbms_lab.employess' doesn't exist

```
Command Prompt - mysql -u X + v

mysql> select * from employess;
ERROR 1146 (42S02): Table 'dbms_lab.employess' doesn't exist
mysql> select * from employees;
+-----+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | manager_id | salary |
+-----+-----+-----+-----+-----+
|      1 | Ramesh   |      10 |      NULL   | 80000.00 |
|      2 | Suresh   |      10 |          1   | 60000.00 |
|      3 | Mahesh   |      20 |          1   | 45000.00 |
|      4 | Kiran    |      30 |          2   | 55000.00 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from customers;
+-----+-----+-----+
| customer_id | customer_name | city |
+-----+-----+-----+
|          201 | Rahul         | Hyderabad |
|          202 | Anita         | Delhi |
|          203 | Kiran         | Mumbai |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from orders;
+-----+-----+-----+-----+
| order_id | customer_id | order_date | total_amount |
+-----+-----+-----+-----+
|      1001 |          201 | 2026-08-01 |      2500.00 |
|      1002 |          202 | 2026-08-02 |      3200.00 |
|      1003 |          201 | 2026-08-03 |      1500.00 |
|      1004 |          203 | 2026-08-04 |      5000.00 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select * from accounts;
```

```
mysql> select * from accounts;
```

account_no	holder_name	balance
101	Bhavya	50000.00
102	Priya	30000.00
103	Arun	45000.00

```
3 rows in set (0.00 sec)
```

```
mysql> select emp_name, dept_name from employee e inner join department d
```

```
->
```

```
-> SELECT emp_name, dept_name
```

```
-> FROM employees e
```

```
-> INNER JOIN departments d
```

```
-> ON e.dept_id = d.dept_id;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right use near 'SELECT emp_name, dept_name
```

```
FROM employees e
```

```
INNER JOIN departments d
```

```
ON e.dept_i' at line 3
```

```
mysql> SELECT emp_name, dept_name
```

```
-> FROM employees e
```

```
-> INNER JOIN departments d
```

```
-> ON e.dept_id = d.dept_id;
```

emp_name	dept_name
Ramesh	CSE
Suresh	CSE
Mahesh	ECE
Kiran	EEE

```
4 rows in set (0.01 sec)
```



```
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	email	role_name
Bhavya	bhavya@gmail.com	Admin
Priya	priya@gmail.com	Manager
Arun	arun@gmail.com	Employee

3 rows in set (0.00 sec)

```
mysql> SELECT name, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	role_name
Bhavya	Admin
Priya	Manager
Arun	Employee

3 rows in set (0.00 sec)

```
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id
->
-> UNION
->
-> SELECT name, email, role_name
-> FROM users u
```



```
mysql> SELECT name, email, role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id = r.role_id
->
-> UNION
->
-> SELECT name, email, role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id = r.role_id;
```

name	email	role_name
Bhavya	bhavya@gmail.com	Admin
Priya	priya@gmail.com	Manager
Arun	arun@gmail.com	Employee

3 rows in set (0.01 sec)

```
mysql> ^C
mysql> SELECT e.emp_name AS Employee,
-> m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id = m.emp_id;
```

Employee	Manager
Ramesh	NULL
Suresh	Ramesh
Mahesh	Ramesh
Kiran	Suresh

4 rows in set (0.00 sec)

```
mysql> SELECT o.order_id,
->         customer_name,
->         city,
->         total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id;
```

order_id	customer_name	city	total_amount
1001	Rahul	Hyderabad	2500.00
1002	Anita	Delhi	3200.00
1003	Rahul	Hyderabad	1500.00
1004	Kiran	Mumbai	5000.00

4 rows in set (0.00 sec)

```
mysql> SELECT dept_name,
->         COUNT(*) AS EmployeeCount,
->         AVG(salary) AS AverageSalary,
->         MAX(salary) AS MaximumSalary
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> GROUP BY dept_name;
```

dept_name	EmployeeCount	AverageSalary	MaximumSalary
CSE	2	70000.000000	80000.00
ECE	1	45000.000000	45000.00
EEE	1	55000.000000	55000.00

3 rows in set (0.01 sec)

```
mysql> SELECT emp_name,
->         salary,
->         dept_name
-> FROM employees e
-> JOIN departments d
```

```
mysql> SELECT emp_name,
->          salary,
->          dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id = d.dept_id
-> WHERE salary > 50000
-> AND dept_name = 'CSE';
```

emp_name	salary	dept_name
Ramesh	80000.00	CSE
Suresh	60000.00	CSE

2 rows in set (0.00 sec)

```
mysql> SELECT customer_name,
->          SUM(total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id = o.customer_id
-> GROUP BY customer_name
-> ORDER BY TotalPurchase DESC;
```

customer_name	TotalPurchase
Kiran	5000.00
Rahul	4000.00
Anita	3200.00

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE faculty(
->     faculty_id INT PRIMARY KEY,
->     faculty_name VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> CREATE TABLE courses(
```

```
mysql> CREATE TABLE faculty(  
->     faculty_id INT PRIMARY KEY,  
->     faculty_name VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> CREATE TABLE courses(  
->     course_id INT PRIMARY KEY,  
->     course_name VARCHAR(50),  
->     faculty_id INT  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> CREATE TABLE students(  
->     student_id INT PRIMARY KEY,  
->     student_name VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> CREATE TABLE enrollments(  
->     student_id INT,  
->     course_id INT  
-> );  
Query OK, 0 rows affected (0.09 sec)  
  
mysql> INSERT INTO faculty VALUES  
-> (1, 'Dr. Kumar'),  
-> (2, 'Dr. Priya');  
Query OK, 2 rows affected (0.01 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> INSERT INTO courses VALUES  
-> (101, 'DBMS', 1),  
-> (102, 'Python', 2);  
Query OK, 2 rows affected (0.01 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> INSERT INTO students VALUES  
-> (1, 'Bhavya'),
```

```
mysql> INSERT INTO students VALUES
-> (1,'Bhavya'),
-> (2,'Priya'),
-> (3,'Arun');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO enrollments VALUES
-> (1,101),
-> (2,102),
-> (3,101);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT student_name,
->         course_name,
->         faculty_name
-> FROM students s
-> JOIN enrollments e
-> ON s.student_id = e.student_id
-> JOIN courses c
-> ON e.course_id = c.course_id
-> JOIN faculty f
-> ON c.faculty_id = f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Arun         | DBMS       | Dr. Kumar   |
| Bhavya       | DBMS       | Dr. Kumar   |
| Priya        | Python     | Dr. Priya   |
+-----+-----+-----+
3 rows in set (0.01 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO accounts
```



```
mysql> INSERT INTO accounts
-> VALUES(104,'Anil',25000);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Bhavya      | 50000.00 |
|          102 | Priya       | 30000.00 |
|          103 | Arun        | 45000.00 |
|          104 | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO accounts
-> VALUES(104,'Anil',25000);
ERROR 1062 (23000): Duplicate entry '104' for key 'accounts.PRIMARY'
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> insert into accounts values(104,'Anil',25000);
ERROR 1062 (23000): Duplicate entry '104' for key 'accounts.PRIMARY'
mysql> ^C
mysql> SELECT * FROM accounts;
```

```
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mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Bhavya      | 50000.00 |
|          102 | Priya       | 30000.00 |
|          103 | Arun        | 45000.00 |
|          104 | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO accounts
    -> VALUES(105,'Ravi',10000);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Bhavya      | 50000.00 |
|          102 | Priya       | 30000.00 |
|          103 | Arun        | 45000.00 |
|          104 | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance - 5000
    -> WHERE account_no = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance + 5000
    -> WHERE account_no = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Bhavya      | 45000.00 |
|          102 | Priya      | 35000.00 |
|          103 | Arun       | 45000.00 |
|          104 | Anil       | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE accounts
    -> SET balance = balance - 3000
    -> WHERE account_no = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
    -> SET balance = balance - 3000
    -> WHERE account_no = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE accounts
    -> SET balance = balance + 3000
    -> WHERE account_no = 999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
|          101 | Bhavya      | 45000.00 |
|          102 | Priya       | 35000.00 |
|          103 | Arun        | 45000.00 |
|          104 | Anil        | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

```